

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring

down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or

40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

2.Question

A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

Answer:

Comprehensive Design and Documentation of Star, Mesh, Bus, and Hybrid Network Topologies

1. Introduction

Network topology defines how networking devices and communication links are physically and logically arranged. The choice of topology affects network performance, scalability, reliability, cost, maintenance, and fault tolerance. This document designs four major network topologies—Star, Mesh, Bus, and Hybrid—using appropriate networking devices, transmission media, cabling standards, and communication protocols.

IEEE 802.3 defines Ethernet physical-layer and MAC-layer operation over media including twisted pair, coaxial cable, and fiber optic cable. The current consolidated IEEE 802.3-2022 standard covers Ethernet technologies from low-speed Ethernet through very high-speed implementations.

2. Star Topology

2.1 Central Networking Device

The central device in a modern star topology is normally an Ethernet switch.

The switch provides a separate physical link between every end device and the central switching point. It learns MAC addresses and forwards unicast frames only toward the appropriate destination port. Broadcast frames, such as ARP requests, are normally forwarded to all ports belonging to the same VLAN, so a switch does not automatically eliminate a broadcast domain; VLANs and Layer-3 interfaces are used to divide broadcast domains.

A hub can also form a star physically, but it repeats traffic to all ports and creates a shared collision domain. Therefore, switches are preferred in modern Ethernet networks.

2.2 Transmission Media and Cabling

Recommended implementation:

Parameter	Specification
Medium	UTP/STP twisted-pair Ethernet
Recommended cable	Cat6 or Cat6A
Connector	RJ45/8P8C

Wiring	TIA-568.2-D structured cabling
Ethernet standard	IEEE 802.3 family
Typical endpoint speed	1 Gbps
10-Gbps option	10GBASE-T
Maximum structured copper channel	100 m
Cat6 at 10 Gbps	Typically up to 55 m
Cat6A at 10 Gbps	Up to 100 m

For 10GBASE-T, Cat6A supports 10 Gbps over up to 100 m, while Cat6 is commonly limited to approximately 55 m for 10-Gbps operation. Cisco documentation also identifies the 100-m channel limit and these Cat6/Cat6A distance considerations.

T568B Example

An RJ45 Ethernet connection can be terminated using the T568B pin arrangement:

1 → 2 → 3 → 4 → 5 → 6 → 7 → 8

White/Orange → Orange → White/Green → Blue → White/Blue → Green
→ White/Brown → Brown

The same wiring convention should normally be used at both ends for a straight-through connection.

2.3 Example Port Assignment

Device	Switch Port	Cable	Maximum Link
PC-01	Fa0/1 or Gi0/1	Cat6	100 m

PC-02	Fa0/2 or Gi0/2	Cat6	100 m
PC-03	Fa0/3 or Gi0/3	Cat6	100 m
Server	Gi0/4	Cat6A	100 m

2.4 Data Flow

Unicast

If PC-01 sends data to PC-03:

PC-01 → Switch → PC-03

The switch examines the destination MAC address and forwards the frame through the appropriate port.

Broadcast

If PC-01 sends an ARP broadcast:

PC-01 → Switch → PC-02, PC-03, Server

The switch forwards the broadcast to ports in the same VLAN.

2.5 Advantages

1. Easy fault isolation: Failure of one cable normally affects only the connected device.
2. Easy expansion: Additional computers can be connected to unused switch ports without redesigning the whole network.

2.6 Disadvantages

3. Central-device dependency: Failure of the main switch can disconnect all attached devices.

4. Higher cabling requirement: Every device requires an individual cable to the central switch, increasing cable length and installation cost.

3. Mesh Topology

3.1 Full Mesh and Partial Mesh

A full mesh connects every node directly to every other node.

For n nodes, the number of required links is:

$$L = n(n-1) / 2$$

The number of interfaces required per node is:

$$P = n - 1$$

Example: For 5 routers:

$$L = \frac{5(5-1)}{2}$$

$$L = 20 / 2 = 10$$

Therefore, 10 direct links are required.

A partial mesh provides redundant connections only between important nodes. It reduces the number of physical links while maintaining alternative paths where reliability is most important.

3.2 When to Use Each Variant

Variant	Appropriate Use
Full mesh	Small critical networks requiring maximum redundancy
Partial mesh	Enterprise core, ISP networks, data centers, banks, WANs

Full mesh	High-cost but extremely fault-tolerant environments
Partial mesh	Large networks where full-mesh cost is impractical

For example, a banking network can use a partial mesh at the core, with multiple core switches, firewalls, routers, and data-center systems connected through redundant paths.

3.3 Required Devices

- Layer-3 switches
- Enterprise routers
- Firewalls
- Servers
- Multiple Ethernet/fiber interfaces
- SFP/SFP+ or higher-speed optical transceivers
- Redundant power supplies

A router participating in a full mesh with five routers would need enough interfaces or logical/physical interfaces to establish the required peer connections.

3.4 Transmission Media

Twisted Pair

Suitable for short-distance connections inside a building. Typical choice: Cat6A, 10GBASE-T, up to 100 m.

Multimode Fiber

Suitable for high-speed connections inside data centers and buildings. Example: OM4 fiber with appropriate optical transceivers.

Single-Mode Fiber

Suitable for long-distance connections between buildings, campuses, or metropolitan locations because it provides much longer reach than copper.

Wireless

Wireless links can be used for geographically separated locations where installing physical cables is difficult, although dedicated fiber is generally preferred for high-reliability core networks.

3.5 Redundancy Protocols

- STP – IEEE 802.1D
- RSTP – IEEE 802.1w
- MSTP – Multiple Spanning Tree
- OSPF
- IS-IS
- BGP
- Link aggregation where appropriate

STP/RSTP prevents Layer-2 switching loops by logically blocking redundant paths until they are needed. Dynamic routing protocols such as OSPF and BGP can select alternative Layer-3 paths when a route becomes unavailable.

3.6 Advantages

5. Excellent fault tolerance: Multiple physical or logical paths allow traffic to continue after individual link failures.

6. High availability: Critical services can remain operational even when equipment or connections fail.

3.7 Disadvantages

7. High cost: More links, interfaces, transceivers, cables, and network equipment are required.
8. Complex management: Routing, redundancy, monitoring, and troubleshooting become more difficult as the number of connections increases.

4. Bus Topology

4.1 Transmission Media

Traditional Ethernet bus networks used coaxial cable.

Standard	Cable	Speed	Maximum Segment
10BASE2	Thin coaxial	10 Mbps	185 m
10BASE5	Thick coaxial	10 Mbps	500 m

10BASE2 commonly used thin coaxial cable such as RG-58-type 50-ohm coaxial cable, while 10BASE5 used thick 50-ohm coaxial cable. BNC connectors were associated with 10BASE2 installations.

IEEE documentation identifies 10BASE5 as a 10-Mbps, 500-m coaxial Ethernet and 10BASE2 as a 10-Mbps, 185-m coaxial Ethernet.

4.2 Physical Layout

4.3 Important Components

Backbone Cable

The backbone is the single shared coaxial transmission medium connecting all stations.

T-Connector

A BNC T-connector provides the physical connection between the network interface and the coaxial backbone.

Terminator

A 50-ohm terminator is placed at both ends of the bus.

The terminators absorb the electrical signal at the end of the cable. Without proper termination, signals can reflect from the cable ends, producing interference and communication errors.

4.4 Signal Propagation

When one computer transmits, the signal propagates along the shared backbone. Multiple devices can observe the transmission.

Because the medium is shared, simultaneous transmissions can result in collisions.

Traditional half-duplex Ethernet used CSMA/CD—Carrier Sense Multiple Access with Collision Detection. IEEE 802.3 specifies CSMA/CD behavior for shared-medium Ethernet.

9. A station listens to the medium.
10. If the medium is idle, it transmits.
11. If another station transmits simultaneously, a collision can occur.
12. The stations detect the collision.

13. Transmission stops.
14. Stations wait for a calculated backoff period.
15. Transmission is attempted again.

Modern switched full-duplex Ethernet normally does not operate with collisions in this way, which is another reason bus Ethernet has become obsolete.

4.5 Advantages

16. Low historical cost: Early bus Ethernet required less dedicated cabling than a separate cable from every device to a switch.
17. Simple physical structure: Devices shared a single backbone cable.

4.6 Disadvantages

18. Backbone failure affects the entire network: A break in the shared cable can disconnect many or all devices.
19. Poor scalability: As more devices share the medium, contention and troubleshooting difficulties increase.

Modern Status

Bus Ethernet such as 10BASE2 and 10BASE5 is primarily of historical/legacy interest. Modern LANs overwhelmingly use switched star or hierarchical star architectures because they provide better performance, scalability, manageability, and fault isolation.

5. Hybrid Topology

5.1 Definition

A hybrid topology combines two or more different basic topologies into one network.

For example, a university can use:

- Star topology inside classrooms
- Star topology inside individual buildings
- Partial mesh between core switches
- Fiber backbone connecting buildings

This creates a practical hierarchical hybrid network.

5.2 Real-World Example: University Campus

Physical Distance Example

- PC → Building Switch: 10–90 m Cat6A
- Building Switch → Core: OM4 fiber
- Building A → Building B: single-mode fiber
- Core switches: redundant fiber links

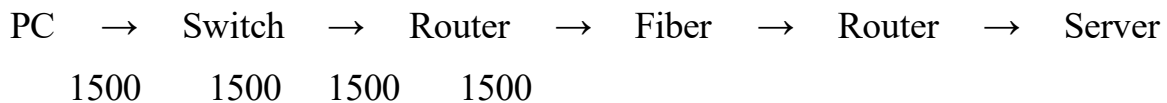
5.3 Devices and Media

Component	Purpose
Layer-2 switches	Connect end devices
Layer-3 switches	Routing between VLANs
Routers	WAN/Internet connectivity
Firewalls	Security and traffic filtering
Access points	Wireless connectivity
Fiber switches	High-speed backbone
Media converters	Convert copper Ethernet to fiber
SFP/SFP+ modules	Optical/electrical interfaces

5.4 Compatibility Between Different Segments

MTU

The Maximum Transmission Unit (MTU) should normally be consistent across Layer-3 paths.



If one segment uses a significantly smaller MTU, fragmentation or Path MTU Discovery problems can occur.

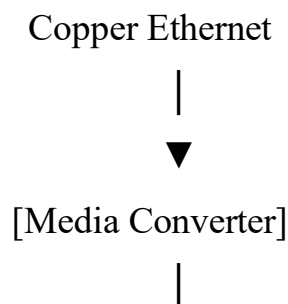
Speed Auto-Negotiation

Ethernet interfaces should be configured to negotiate compatible speeds where supported.



For critical links, fixed speed/duplex settings may be used when required by the equipment design, but mismatched manual settings should be avoided.

Media Conversion





Fiber Ethernet

The converter changes the physical medium, not necessarily the higher-layer protocol.

Protocol Translation

If two segments use fundamentally different communication protocols, a gateway or protocol translator may be necessary. For ordinary Ethernet-to-fiber conversion, however, a media converter is sufficient because Ethernet can remain unchanged at the higher layers.

5.5 Advantages

- 20. Design flexibility: Different areas can use different topologies according to their requirements.
- 21. Scalability: New buildings, departments, or network segments can be added without redesigning the entire organization.

5.6 Disadvantages

- 22. Management complexity: Multiple technologies require more planning, monitoring, documentation, and troubleshooting.
- 23. Higher implementation cost: Different switches, routers, fiber modules, converters, and cabling systems may be required.

6. Overall Comparison

Factor	Star	Mesh	Bus	Hybrid
Basic structure	Central switch	Multiple interconnected	Shared backbone	Combination

		nodes		
Scalability	High	Low–medium for full mesh; high for partial mesh	Low	Very high
Expansion	Add switch port	Add links/interfaces	Extend backbone carefully	Add a new topology segment
Fault tolerance	Medium	Very high	Low	Medium–very high
Main failure point	Central switch	Few/no single links in full mesh	Backbone	Depends on design
Recovery	Replace/fail over switch	Alternate path	Usually requires backbone repair	Redundant segment/path
Cabling cost	Medium	Very high	Low historically	Medium–high
Management	Easy	Complex	Simple conceptually, difficult troubleshooting	Complex
Performance	High	Very high	Low under	High if well

			contention	designed
Typical media	Cat6/Cat6A	Fiber/Cat6A	Coaxial	Copper + fiber + wireless
Typical use	Offices, labs, homes	Data centers, WAN/core	Legacy networks	Campuses, enterprises
Typical size	Small–large	Small critical core to large WAN	Small legacy networks	Medium–large

7. Scalability and Node Considerations

There is no single universal "maximum node count" for a topology. Practical limits depend on the Ethernet standard, switch port density, VLAN design, routing architecture, addressing, bandwidth, and application requirements.

Star

A 48-port switch can directly accommodate approximately 48 connected endpoints, subject to reserved uplink ports and design requirements. Larger networks are created by connecting multiple switches hierarchically.

Full Mesh

For n nodes:

$$\text{Connections} = n(n-1) / 2$$

Nodes	Full-Mesh Links
4	6

5	10
10	45
20	190
50	1,225

Therefore, full mesh becomes expensive as the number of nodes increases. Partial mesh is more practical for large networks.

Bus

The number of nodes is limited by the particular legacy Ethernet specification, cable length, repeaters, signal quality, and collision-domain constraints. The important limitation is that all nodes share the same medium.

Hybrid

Hybrid networks can scale to very large organizations by dividing the infrastructure into access, distribution, and core layers.

8. Practical Design Decision

Scenario

Consider an organization with:

- 500 employees
- Three buildings
- Central data center
- Internet connectivity
- VoIP phones
- Wi-Fi

- Servers
- High availability requirements

A suitable architecture would

This is a hybrid topology:

- Star at the access layer
- Partial mesh at the core
- Fiber backbone between buildings
- Cat6A for endpoint connections
- Layer-3 routing between major network segments
- Redundant core links for high availability

9. Recommendation Framework

Requirement	Recommended Topology
Small budget and simple LAN	Star
Maximum redundancy for a small critical network	Full Mesh
High availability enterprise core	Partial Mesh
Legacy/simple shared network	Bus, only where legacy compatibility requires it
Large campus or enterprise	Hybrid
Easy expansion	Star/Hybrid
Very high fault tolerance	Mesh/Hybrid
Lowest modern maintenance	Star

complexity	
Multiple buildings	Hybrid with fiber backbone
Critical financial/data-center infrastructure	Partial Mesh/Hybrid

Final Recommendation

For a small office or computer laboratory, a switched star topology is generally the best choice because it is inexpensive, easy to install, and simple to troubleshoot.

For a bank, data center, or other mission-critical environment, a partial mesh should be used for the core because redundant paths provide higher availability.

For a large university or enterprise campus, a hybrid topology is usually the strongest practical solution because it combines star access networks with redundant fiber-based core connections.

A traditional bus topology should generally be avoided for new installations because its shared-medium architecture, collision behavior, limited scalability, and legacy coaxial cabling make it inferior to modern switched Ethernet.

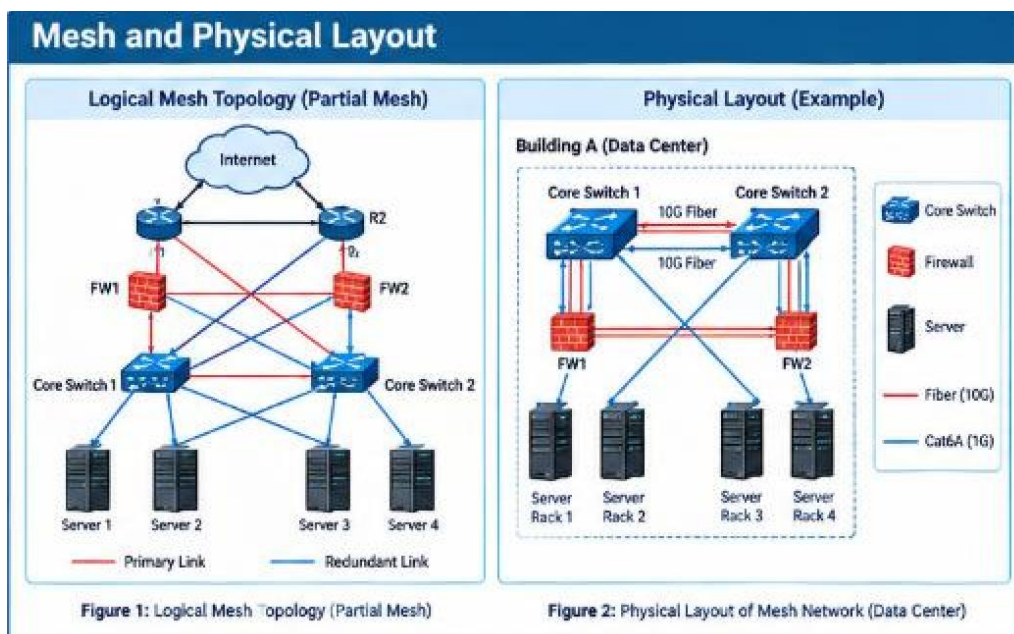
10. Conclusion

Star, mesh, bus, and hybrid topologies provide different solutions to networking requirements. Star topology provides simplicity and easy expansion, while mesh topology provides superior redundancy and fault tolerance. Bus topology is historically important because it introduced shared-medium Ethernet but is now largely obsolete for modern LAN deployment. Hybrid topology combines the strengths of multiple architectures and is therefore particularly suitable for large campuses and enterprises.

From a BL2 understanding perspective, the topologies demonstrate how device placement, transmission media, addressing, switching, routing, and redundancy influence network behavior. From a BL6 creating perspective, a practical enterprise can combine switched star access networks, redundant mesh core links, fiber backbones, and appropriate routing protocols to create a scalable and fault-tolerant network.

The final selection should therefore consider organizational size, available budget, reliability requirements, physical distance, expected traffic, security requirements, and future growth rather than choosing a topology solely on the basis of its theoretical structure.

DIAGRAM:



4.2 Physical Layout

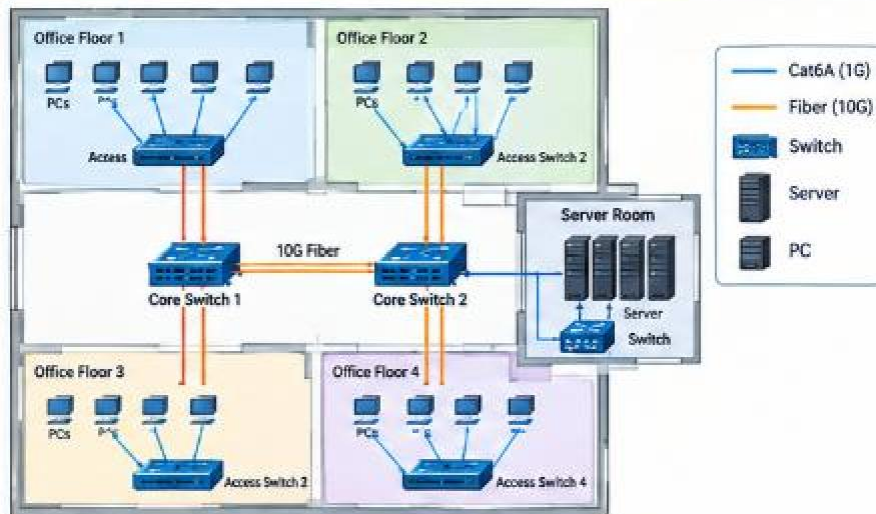


Figure 3: Physical Layout of a Multi-Floor Office Network

5.2 Real World Example – University Campus Network

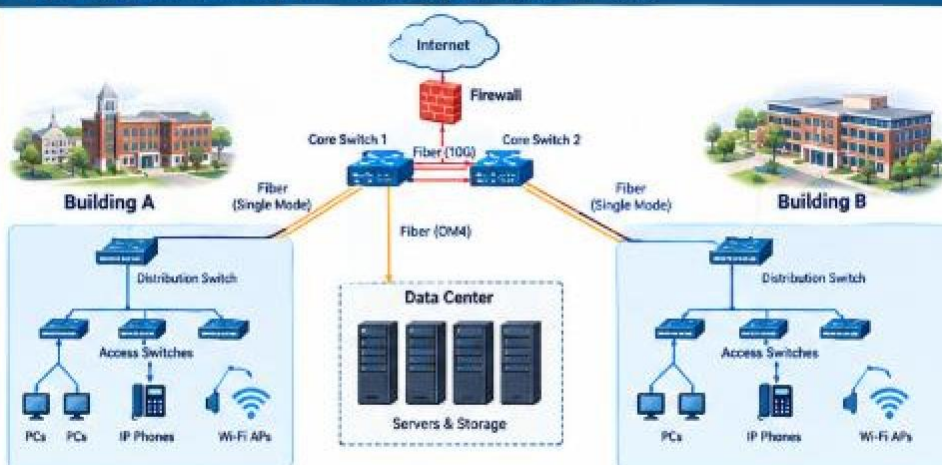
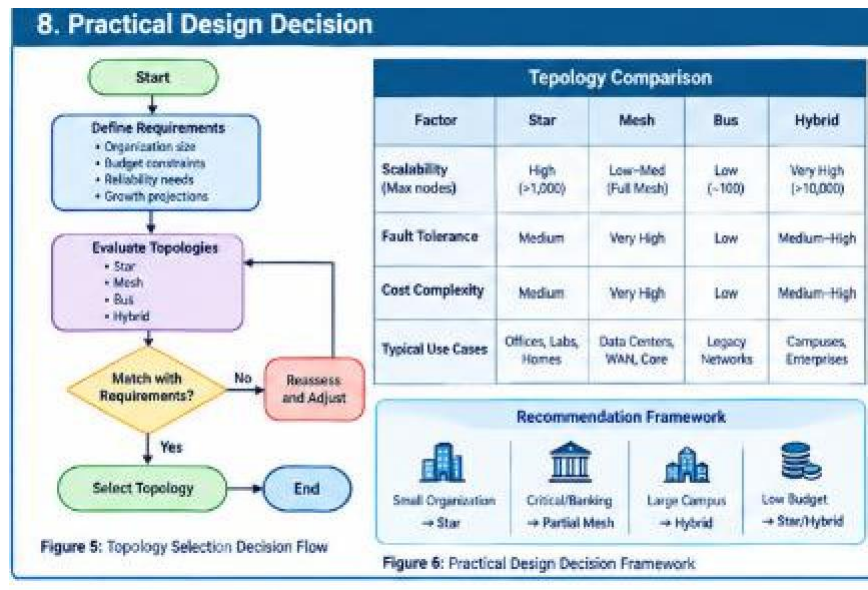


Figure 4: Real-World Example – University Campus Network



TOOLS USED:

CISCO PACKET TRACER AND ROVO IMAGE TOOL,PYTHON

PROGRAM:

```

def create_mesh(n):

    connections = []

    for i in range(n):

        for j in range(i + 1, n):

            connections.append((f"Router-{i+1}", f"Router-{j+1}"))

    return connections

n = int(input("Enter the number of routers: "))
        
```

```

if n < 2:

    print("Enter at least 2 routers.")

else:

    connections = create_mesh(n)


    print("\nFULL MESH TOPOLOGY")

    print("-----")


    for a, b in connections:

        print(f"{a} <---- Fiber ----> {b}")


    total = len(connections)

    expected = n * (n - 1) // 2


    print("\nNumber of Routers:", n)

    print("Required Connections:", total)

    print("Formula: n(n-1)/2")

    print("Calculated Connections:", expected)


    if total == expected:

        print("Verification: All routers are directly connected.")

        print("Mesh topology created successfully.")

```

```
Enter the number of routers: 4

FULL MESH TOPOLOGY
-----
Router-1 <---- Fiber ----> Router-2
Router-1 <---- Fiber ----> Router-3
Router-1 <---- Fiber ----> Router-4
Router-2 <---- Fiber ----> Router-3
Router-2 <---- Fiber ----> Router-4
Router-3 <---- Fiber ----> Router-4

Number of Routers: 4
Required Connections: 6
Formula:  $n(n-1)/2$ 
Calculated Connections: 6
Verification: All routers are directly connected.
Mesh topology created successfully.

...Program finished with exit code 0
Press ENTER to exit console.
```

References

1. IEEE, IEEE Standard for Ethernet, IEEE Std 802.3-2022. IEEE 802.3 defines Ethernet operation across coaxial, twisted-pair, fiber-optic, and other media.
2. IEEE, IEEE 802.3a—10BASE2 Ethernet, IEEE Standards. The 10BASE2 implementation uses thin coaxial cable and supports a 185 m segment.
3. IEEE, IEEE 802.3an—10GBASE-T, IEEE Standards. 10GBASE-T specifies 10-Gbps Ethernet over balanced twisted-pair structured cabling up to 100 m.
4. Cisco, Catalyst 9300 Series Hardware Installation Guide, Cisco Systems. Provides practical Ethernet cabling and 10GBASE-T reach information for Cat6 and Cat6A.
5. Cisco, Ethernet 100BASE-TX and 10BASE-T Physical Characteristics, Cisco Systems. Provides Ethernet segment-length and physical-medium information.